ClipSAT : Contest Math & Olympiad AMC 10/12, Kangaroo & Mawhiba

Unit 5.01: Recurrence Relations, Markov Chains & Geometric Probability

Name _____

Q1. The coordinate diagram shows the parabolic curve f_1x = x^2 - 14x + 64 in the domain of Recurrence Relations & Geometric Probability. By completing the square f_1x = x - h^2 + k , find the sum k + f_19 .

[ ] A: 34
[ ] B: 41
[ ] C: 19
[ ] D: 32

Q2. A mathematical model in Recurrence Relations & Geometric Probability is defined by the cubic function y_2x = 3x^3 - 7x^2 + 15x - 12 . Calculate the exact gradient of the tangent to the curve at the point where x = 3 .

[ ] A: 96
[ ] B: 54
[ ] C: 39
[ ] D: 28

Q3. In Recurrence Relations & Geometric Probability, two displacement vectors are given by mathbf{u}_{3} = langle 7, -8 rangle and mathbf{v}_{3} = langle 9, 7 rangle . Compute the exact scalar value of mathbf{u}_{3} cdot mathbf{v}_{3} + |mathbf{u}_{3}|^2 .

[ ] A: 7
[ ] B: 113
[ ] C: 120
[ ] D: 232

Q4. The linear growth profile shown in the diagram for Recurrence Relations & Geometric Probability follows an arithmetic progression with first term a_1 = 8 and common difference d = 6 . Find the exact sum S_{15} of the first 15 terms.

[ ] A: 92
[ ] B: 840
[ ] C: 1380
[ ] D: 750

Q5. In Recurrence Relations & Geometric Probability, evaluate the exact definite integral I_5 = int_0^{2} 18x^2 + 28x,mathrm{d}x .

[ ] A: 104
[ ] B: 128
[ ] C: 132
[ ] D: 53

Q6. In Recurrence Relations & Geometric Probability, consider the quadratic function f_6x = x^2 - 16x + 151 . Express f_6x in the completed-square form x - h^2 + k and compute the exact value of k + f_615 .

[ ] A: 231
[ ] B: 223
[ ] C: 136
[ ] D: 216

Q7. The graph illustrates the cubic model y_7x = 3x^3 - 8x^2 + 33x - 19 studied in Recurrence Relations & Geometric Probability. Find the instantaneous gradient frac{mathrm{d}y}{mathrm{d}x} of the tangent to the curve at x = 8 .

[ ] A: 609

[ ] B: 448

[ ] C: 481

[ ] D: 89

Q8. In Recurrence Relations & Geometric Probability, two displacement vectors are given by mathbf{u}_{8} = langle 8, -7 rangle and mathbf{v}_{8} = langle 8, 5 rangle . Compute the exact scalar value of mathbf{u}_{8} cdot mathbf{v}_{8} + |mathbf{u}_{8}|^2 .

[ ] A: 29

[ ] B: 113

[ ] C: 212

[ ] D: 142